CESM1.0.5 NAHosMIP Annual-Mean Archive

Final Data Structure — Common 100-Year Window (2051–2150)

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1 Overview

This document describes the final, verified directory structure of the annual-mean NetCDF archive under `CESM1.0.5_Model_Runs_Outputs_Annual_Means/`, after re-extraction restricted every one of the 27 member simulations and all 9 ensemble-mean experiments to the single 100-year period common to all of them: **2051–2150**. Coverage was verified file-by-file (filename year range and, for a spot-check, the embedded NetCDF time axis) with zero gaps, zero off-window files, and no duplicate vintages remaining.

The 27 simulations are a 3 members $\times$ 9 experiments design: 3 CO_2 levels ($1\times$, $2\times$, $4\times$) $\times$ 3 North Atlantic hosing treatments ($+0.3$ Sv, -0.3 Sv, no hosing), run three times with different round-off/restart perturbations to isolate internal variability (Member 1: `BFBFLAG=FALSE`; Member 2: `BFBFLAG=TRUE`; Member 3: `BFBFLAG=TRUE` plus an ocean temperature round-off perturbation).

2 Top-Level Directory Layout

```
CESM1.0.5_Model_Runs_Outputs_Annual_Means/
|-- 00_EXTRACTION_METHODOLOGY.{tex,pdf,...} (original pipeline design doc)
|-- 01_DATA_STRUCTURE_2051-2150.{tex,pdf} (this document)
|-- scripts/ (pipeline code + logs)
|
|-- Member1_BFBFLAG-FALSE/ 9 cases, 771 files total
|-- Member2_BFBFLAG-TRUE/ 9 cases, 819 files total
|-- Member3_OceanTempPerturbation/ 9 cases, 819 files total
`-- EnsembleMean_3Member/ 9 cases, 723 files total
```

Each member directory contains one subdirectory per experiment case, and each case directory contains an `annual/` folder (every variable, every component) and, for cases with daily atmosphere output, a sibling `daily/` folder (full daily-resolution series, 8 variables, no temporal averaging):

```
<Member>/<case_name>/
|-- annual/
| |-- <case>.<VAR>.annual.2051-2150.nc
| `-- <case>.<VAR>_daily.annual.2051-2150.nc (atm_day-derived vars only)
`-- daily/
    `-- <case>.<VAR>.daily.2051-2150.nc (atm_day vars only)

EnsembleMean_3Member/<case>_EnsembleMean/
`-- annual/
    `-- <case>_EnsembleMean.<VAR>.annual.2051-2150.nc
```

Filename convention: every file is `<output_case_name>.<VARNAME[_daily]>.<annual|daily>.2051-2150.nc` — one file per variable, self-describing. After this extraction pass, *every* file in the archive carries exactly this `2051-2150` tag; no other year range appears anywhere in the tree.

3 Cases per Member

All three members share the same 9 experiment names (hosing treatment × CO_2 level); only the output-directory suffix differs by member.

Experiment	Case directory name per member
1×CO_2, +0.3 Sv hosing	`B1850CN_hos0p3Sv_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation]`
1×CO_2, −0.3 Sv hosing	`B1850CN_neghos0p3Sv_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation]`
1×CO_2, no hosing	`B1850CN_nohosing_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation]`
2×CO_2, +0.3 Sv hosing	`B1850CN_2xCO2_hos0p3Sv_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation`
2×CO_2, −0.3 Sv hosing	`B1850CN_2xCO2_neghos0p3Sv_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbat`
2×CO_2, no hosing	`B1850CN_2xCO2_nohosing_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation`
4×CO_2, +0.3 Sv hosing	`B1850CN_4xCO2_hos0p3Sv_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation`
4×CO_2, −0.3 Sv hosing	`B1850CN_4xCO2_neghos0p3Sv_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbat`
4×CO_2, no hosing	`B1850CN_4xCO2_nohosing_f19g16_NAHosMIP_v5[_BFBT|_OceanTempPerturbation`

Member 1 case directories carry no suffix (plain name as shown); Member 2 appends `_BFBT`; Member 3 appends `_OceanTempPerturbation`. Ensemble-mean case directories use the Member-1 (unsuffixed) name with an `_EnsembleMean` suffix.

4 Variable Inventory

4.1 Atmosphere, monthly (CAM `h0`) — 40 variables, annual mean

```
TREFHT, TREFMNAV, TREFMXAV, QREFHT, RHREFHT, TMQ, QFLX, PRECC, PRECL,
PRECSC, PRECSH, PRECSL, PRECT, SNOWHICE, SNOWHLND, CLDHGH, CLDLOW, CLDMED,
CLDTOT, LWCF, SWCF, FLDS, FLDSC, FLNS, FLNSC, FLNT, FLNTC, FLUT, FLUTC,
FSDS, FSDSC, FSNS, FSNSC, FSNT, FSNTC, LHFLX, SHFLX, SOLIN, SRFRAD, ICEFRAC
```

4.2 Atmosphere, daily (CAM `h1`) — 8 variables

```
TREFHT, TREFHTMX, TREFHTMN, TS, PRECT, PRECL, PRECC, PRECSH
```

Written twice: as an annual mean of the daily series (`annual/`, `_daily` suffix, e.g. `PRECT_daily`) and as the full daily-resolution series (`daily/`, no suffix needed).

4.3 Ocean (POP2 `pop.h`) — 12 variables, annual mean

```
SALT, TEMP, SSH, HMXL, SFWF, SALT_F, SHF, IFRAC, BSF (cdo path)
MOC, N_HEAT, N_SALT (NCO path,
                                                    native dims kept)
```

4.4 Sea ice (CICE `cice.h`) — up to 12 variables, annual mean

```
aice, hi, hs, Tsfc, uvel, vvel, fresh, fsalt, fhocn, strairx, strairy, albsni
```

4.5 Land (CLM `clm2.h0`) — up to 15 variables, annual mean

```
TREFMXAV, TREFMNAV, RAIN, SNOW, QRUNOFF, QOVER, TWS, SOILLIQ, SOILICE,
SNOWDP, FSNO, H2OSNO, TLAI, GPP, NPP
```

5 Per-Case Variable Counts

Case type	Variables	Notes
Member 1/2/3: hosing or neg.-hosing cases (6 of 9 per member)	83 annual + 8 daily = 91 files	Full component set, daily atm. output present
Member 2/3: nohosing cases (3 of 9 per member)	83 annual + 8 daily = 91 files	Daily atm. output present (standard case archive)
Member 1: nohosing cases (3 of 9)	75 annual, no daily	Sourced from archive-only/control-run directories that never archived `cam2.h1`; documented, not a gap
Ensemble mean: hosing/neg.-hosing experiments (6 of 9)	83 annual	No daily/ output for ensemble means (annual only, by design)
Ensemble mean: nohosing experiments (3 of 9)	75 annual	Inherits Member 1's missing-daily nohosing cases

`albsni` (ice) and `TWS` (land) are dropped automatically, per case, wherever genuinely absent from the source history files (logged, not a failure); this is why some cases show slightly fewer than the nominal 87-variable maximum.

6 Provenance Note

This structure reflects a full Stage A/B/C re-extraction (all 27 cases, all 9 ensemble-mean experiments) run on 2026-11-05/06, restricted to the 2051–2150 window common to every raw simulation. A targeted follow-up re-run of the atmosphere-daily component corrected a date-parsing defect in `lib_common.sh/stage_a_merge.sh`: daily (`cam2.h1`) history files are named `...YYYY-MM-DD-SSSSS.nc` rather than the monthly `...YYYY-MM.nc` pattern the original year-extraction logic assumed, which silently mis-computed the daily component's year range for every case. The fix and the full re-run are reflected in every file in this archive; stale pre-fix files were identified and removed as part of the same pass.